Machine Learning with Graphs: Homework II

Dev Sanghvi - ds221

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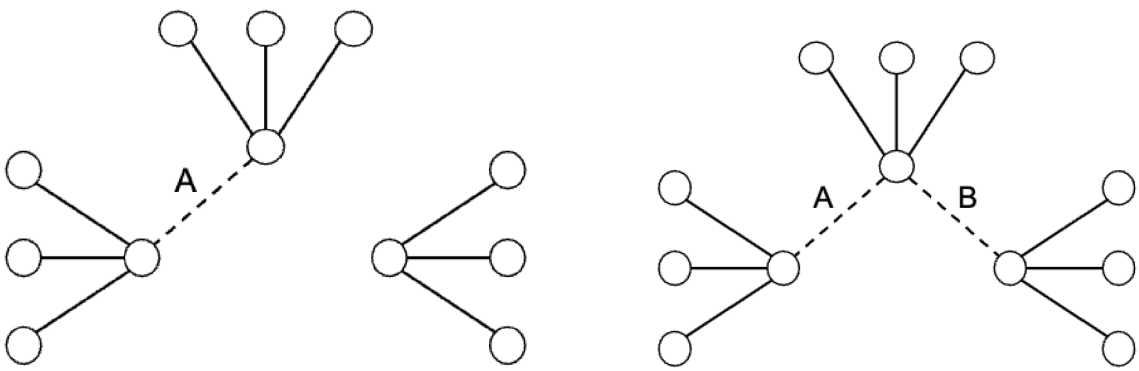
Problem 1



(a) A is a true missing edge

(b) A and B are true missing edges

Figure 1: Problem 1 graphs.

Consider *preferential attachment* score for link prediction, where an edge is predicted based on the degrees of its endpoints. The probability $p(u, v)$ of the edge (u, v) is given as follows:

$$p(u, v) \propto \deg(u) \deg(v)$$

Task. Compute the AUC for preferential attachment using the graph from Figure 1a, where A is the only true missing edge. Do the same for the graph from Figure 1b, where both A and B are true missing edges. Do these values reflect the difference in performance?

Solution. Based on the visual representation, we model the graph as three disconnected star components centered at hubs L, T, and R, each with 3 leaf nodes. The Preferential Attachment (PA) score for a potential edge (u, v) is $PA(u, v) = \deg(u) \times \deg(v)$.

- Hub-Hub edges (e.g., $L - T$, $T - R$, $L - R$) have score $3 \times 3 = 9$.
- Hub-Leaf edges have score $3 \times 1 = 3$.
- Leaf-Leaf edges have score $1 \times 1 = 1$.

Evaluation Setup: We define the candidate set for AUC calculation as *all non-existing edges* in the graph. The goal is to rank the true missing edges (positive set) higher than all other non-edges (negative set).

Case (a): Target set is $\{A = (L, T)\}$. The target edge A (Positive) has score $3 \times 3 = 9$. We compare this against all 56 Negative non-edges.

- **Negatives with Score 9:** There are 2 such edges: (L, R) and (T, R). Since Score(Pos) = Score(Neg), these count as 0.5 correct each. Contribution: $2 \times 0.5 = 1.0$.
- **Negatives with Score 3:** There are 18 such edges (Hub-Leaf non-edges). $9 > 3$, so these are correctly ranked. Contribution: $18 \times 1.0 = 18.0$.
- **Negatives with Score 1:** There are 36 such edges (Leaf-Leaf non-edges). $9 > 1$, so these are correctly ranked. Contribution: $36 \times 1.0 = 36.0$.

Total Correct Pairs $= 1.0 + 18.0 + 36.0 = 55.0$.

$$AUC = \frac{\text{Total Correct}}{\text{Total Negatives}} = \frac{55}{56} \approx 0.9821$$

Calculated AUC: 0.9821

Case (b): Target set is $\{A = (L, T), B = (T, R)\}$. We have 2 Positive edges (A, B) both with Score 9. We compare these against the remaining $57 - 2 = 55$ Negative non-edges.

- **Negatives with Score 9:** There is 1 such edge: (L, R). Comparison with 2 Positives: $2 \times 0.5 = 1.0$ (Ties).
- **Negatives with Score 3:** There are 18 such edges. Comparison with 2 Positives: $2 \times 18 \times 1.0 = 36.0$ (Wins).
- **Negatives with Score 1:** There are 36 such edges. Comparison with 2 Positives: $2 \times 36 \times 1.0 = 72.0$ (Wins).

Total Correct Pairs $= 1.0 + 36.0 + 72.0 = 109.0$. Total Pairs $= 2 \times 55 = 110$.

$$AUC = \frac{109}{110} = 0.990909\cdots \approx 0.9909$$

Calculated AUC: 0.9909

Conclusion. The AUC values (0.9821 vs 0.9909) do reflect a slight difference, but they are misleadingly high. The metric is dominated by the vast majority of easy-to-classify negative edges (e.g., Leaf-Leaf pairs with low degree product). In Case (a), the true missing edge $A = (L, T)$ is tied for first place with two false positives ((L, R) and (T, R)). A strict top-k ranking metric would reveal this ambiguity better than AUC. Therefore, a better alternative would be Hits@k (with $k = 3$) or Mean Reciprocal Rank (MRR), which focus on the rank of the true positive among the highest-scoring candidates rather than the global discrimination against all negatives.

Comparison with Common Neighbors. Common Neighbors (CN) fails in this scenario. The following calculation uses the setup from Case (a) (1 Positive, 56 Negatives), though the result is similar for Case (b). Total Negatives = 56.

- **False Positives (Score 1):** Any pair of leaves attached to the same hub shares that hub as a common neighbor. There are $\binom{3}{2} \times 3 = 9$ such pairs. Since $1 > 0$ (True Positive Score), these are ranked incorrectly.

- **True Positives (Score 0):** The missing edges connect two different hubs (e.g., L and T), which share no common neighbors ($CN = 0$).
- **True Negatives (Score 0):** The remaining $56 - 9 = 47$ non-edges have score 0. These are tied with the True Positive.
- **AUC Calculation:**

$$AUC = \frac{0.0 \times 9 + 0.5 \times 47}{56} = \frac{23.5}{56} \approx 0.4196$$

Since $AUC < 0.5$, the CN predictor is performing worse than random guessing for ranking the missing links, making PA the superior choice.

Problem 2 (optional)

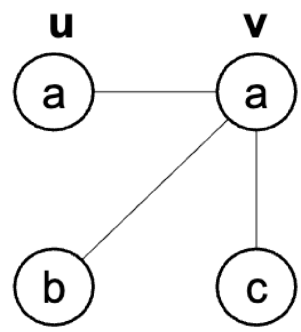


Figure 2: Problem 2 multiset example.

Consider a node classification model for attributed graphs where each node has a single attribute. For each node v, we collect the multiset of attributes within one hop from v (i.e., v and its neighbors). We generate vector representations in $\mathbb{R}^D$ for each vertex to represent its multiset. Use ℓ for possible attributes.

(a) Number of possible multisets

Question. Given ℓ possible attributes, what is the number of possible multisets for a node with degree d?

Solution. This is a problem of counting combinations with replacement (multisets). For a node v with degree d, the multiset contains $k = d + 1$ attributes (itself plus d neighbors), drawn from ℓ possible types. The number of such multisets is given by the multiset coefficient:

$$M_d = \binom{\ell + k - 1}{k} = \binom{\ell + d}{d + 1}$$

(b) Dimensions for 32-bit representation

Question. Assuming that each floating point in the (embedding) vector has 32 bits, how many dimensions are needed to represent these multisets?

Solution. To uniquely represent M_d distinct multisets, we need $\lceil \log_2(M_d) \rceil$ bits of information. Since each floating point number has 32 bits, the number of dimensions D required is given by the total bits divided by 32, rounded up:

$$D = \left\lceil \frac{\log_2(M_d)}{32} \right\rceil = \left\lceil \frac{\log_2 \binom{\ell+d}{d+1}}{32} \right\rceil$$

(c) Barabasi-Albert Model Simulation

Question. If the graph is generated using the Barabasi-Albert model with each new node connecting to $c = 5$ existing nodes, $n \to \infty$, and $\ell = 100$, what would be a good estimate of the

expected number of dimensions needed to represent at least half of the nodes? And to represent 99% of the nodes? And to represent all the nodes?

Solution. Since $n \to \infty$, we must use the asymptotic degree distribution of the Barabasi-Albert model with parameter $m = 5$. The degree distribution follows a power law with exponent $\gamma = 3$:

$$P(k) = \frac{2m(m+1)}{k(k+1)(k+2)}$$

For $m = 5$, this becomes $P(k) = \frac{60}{k(k+1)(k+2)}$. Based on the cumulative distribution function (CDF), we determine the degree thresholds.

- **For 50% of the nodes:** We sum $P(k)$ starting from $k = m$ until the cumulative probability exceeds 0.5.

$$P(5) = \frac{60}{5(6)(7)} = \frac{60}{210} \approx 0.2857$$
$$P(6) = \frac{60}{6(7)(8)} = \frac{60}{336} \approx 0.1786$$
$$P(7) = \frac{60}{7(8)(9)} = \frac{60}{504} \approx 0.1190$$

  Sum: $0.2857 + 0.1786 + 0.1190 = 0.5833 > 0.5$. Thus, the threshold is $d_{50\%} = 7$. The number of possible multisets for $d = 7$ is $M_7 = \binom{100+7}{8} \approx 3.26 \times 10^{11}$. Dimensions required:

$$D_{50\%} = \left\lceil \frac{\log_2(3.26 \times 10^{11})}{32} \right\rceil = \left\lceil \frac{38.2}{32} \right\rceil = 2$$

- **For 99% of the nodes:** We need $\sum_{k=5}^{K} P(k) \geq 0.99$, which means the tail probability $\sum_{k=K+1}^{\infty} P(k) \leq 0.01$. Using the continuous approximation for the tail of the power law $P(k) \approx 2m^2k^{-3}$:

$$\int_K^\infty 2m^2x^{-3}dx = \left[-m^2x^{-2}\right]_K^\infty = \frac{m^2}{K^2}$$

  Setting $\frac{25}{K^2} \leq 0.01 \Rightarrow K^2 \geq 2500 \Rightarrow K \geq 50$. Exact numerical summation of the formula $P(k) = \frac{60}{k(k+1)(k+2)}$ confirms that $\sum_{k=5}^{54} P(k) \approx 0.9903 > 0.99$. Thus, $d_{99\%} = 54$.

$$M_{54} = \binom{100+54}{55} \approx 2.61 \times 10^{42}$$

  Dimensions required:

$$D_{99\%} = \left\lceil \frac{\log_2(2.61 \times 10^{42})}{32} \right\rceil = \left\lceil \frac{140.9}{32} \right\rceil = 5$$

- **For all the nodes:** As $n \to \infty$, the maximum degree $d_{\max}$ scales as $d_{\max} \sim n^{1/(\gamma-1)} = n^{1/2}$ (derived from $\int_{d_{\max}}^{\infty} P(k)dk \approx 1/n$). Consequently, M_d grows without bound, and the dimensions $D \sim \log(M_d)$ grow logarithmically. No finite D suffices for the infinite limit, but for any finite n, D is small.